

Indian Institute Of Technology Gandhinagar



Assignment Report

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1 Question

Design an Operational Transconductance Amplifier (OTA) for the following specifications.

1. Differential Gain = 40 - 60 dB (Gain should vary between 40 to 60 dB by changing V_{cont})
2. CMRR > 70dB
3. ICMR: 0.80 V - 1.4 V
4. Output Swing > 800m V
5. Power < 20 uW
6. Load Capacitance = 1pF

Perform frequency analysis and calculate bandwidth of the designed amplifier. Plot the frequency response (both Gain and Phase) with the capacitive load.

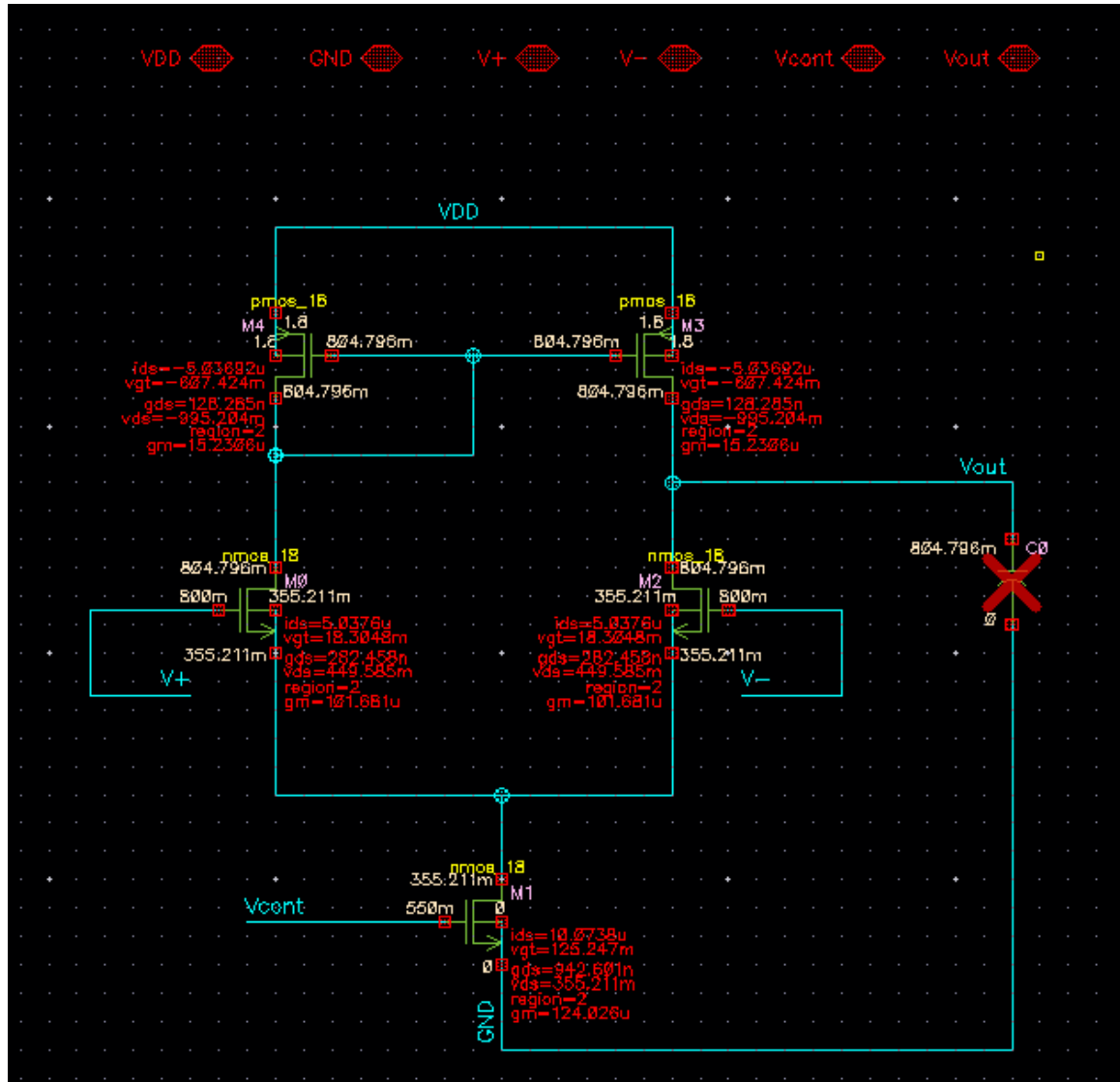


Figure 1: Schematic of single stage Op-Amp

Observations

	Width(uM)	length(uM)	Region	gm(uS)	gds(uS)	Vov(mV)
M1=M2	1.2	2	2	101.68Us	282.458uS	
M4	3.8	1	2	124.026	942.6nS	125.24mV
M5=M6	10	0.95	2	15.23uS	128.285uS	607.424mV
Vdc	1.8v	Vbias(tail)	0.5V	Iss	10uA	
A(dm)	48dB	A(cm)		CMRR		Output swing=830mV

Table 1: table representing various parameters

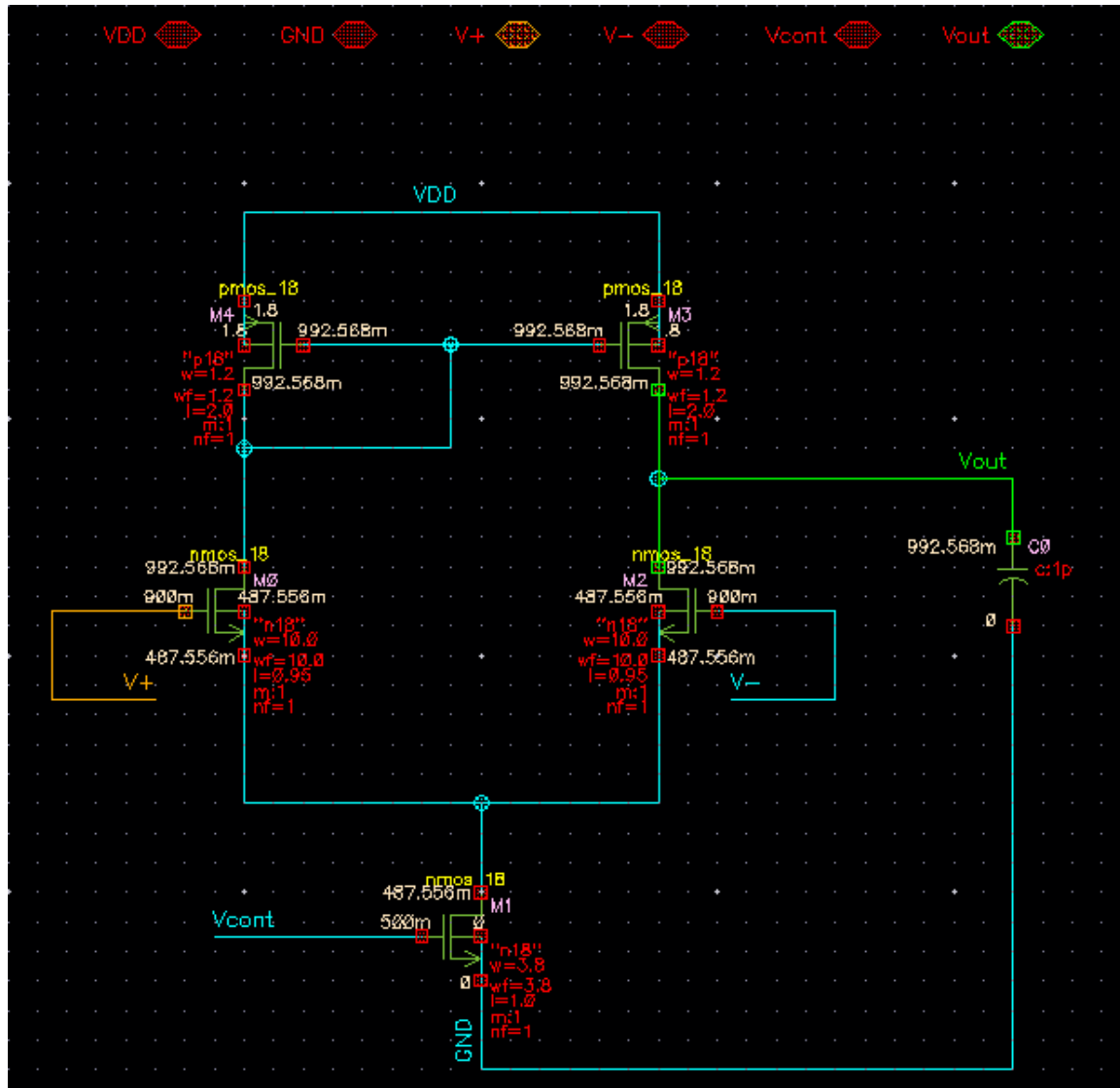


Figure 2: Schematic of single stage Op-Amp with width and length

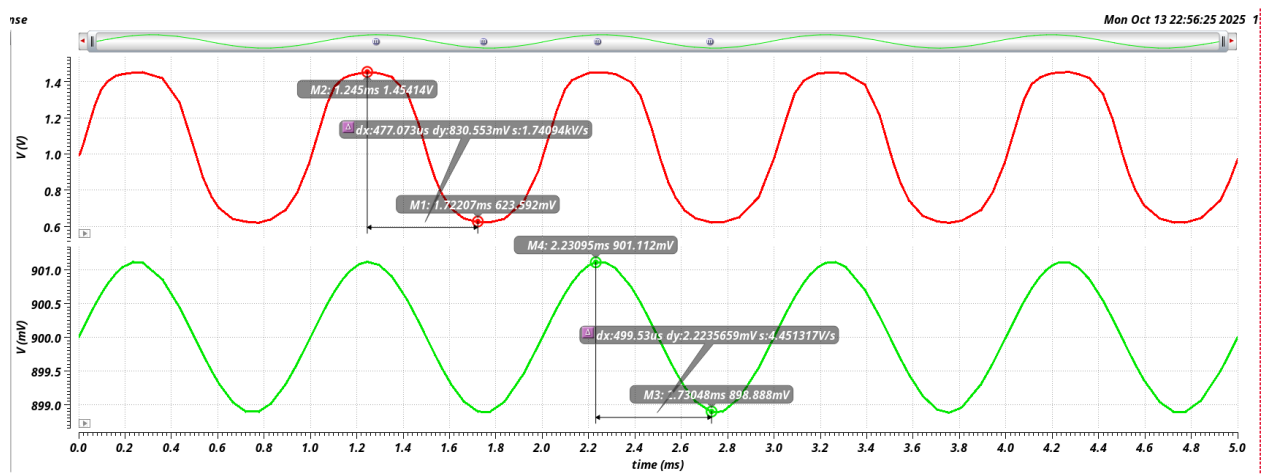


Figure 3: output swing

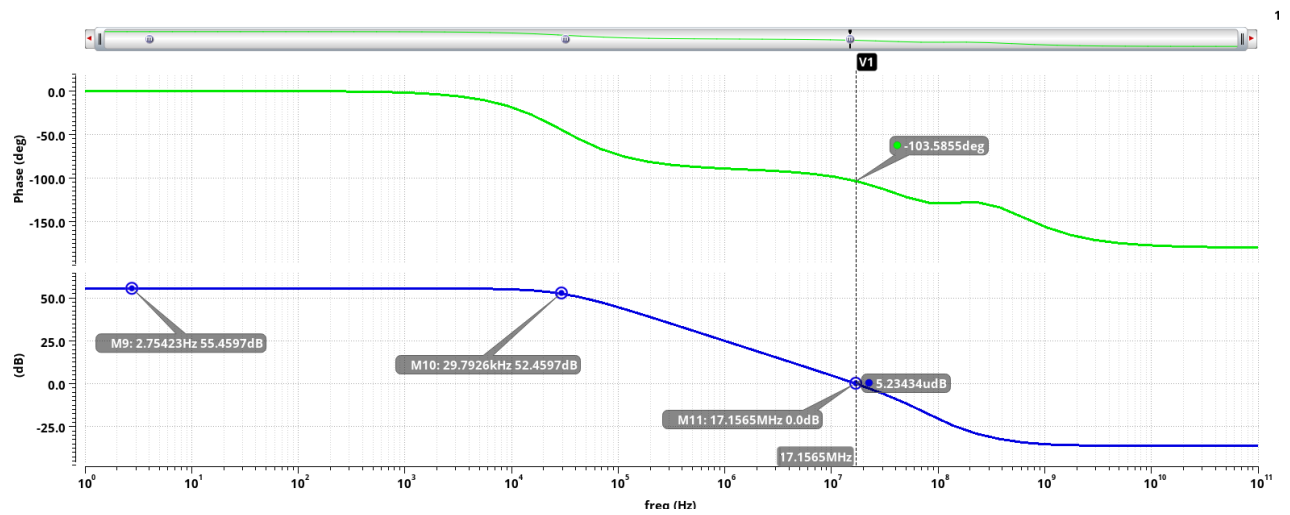


Figure 4: phase and magnitude plot

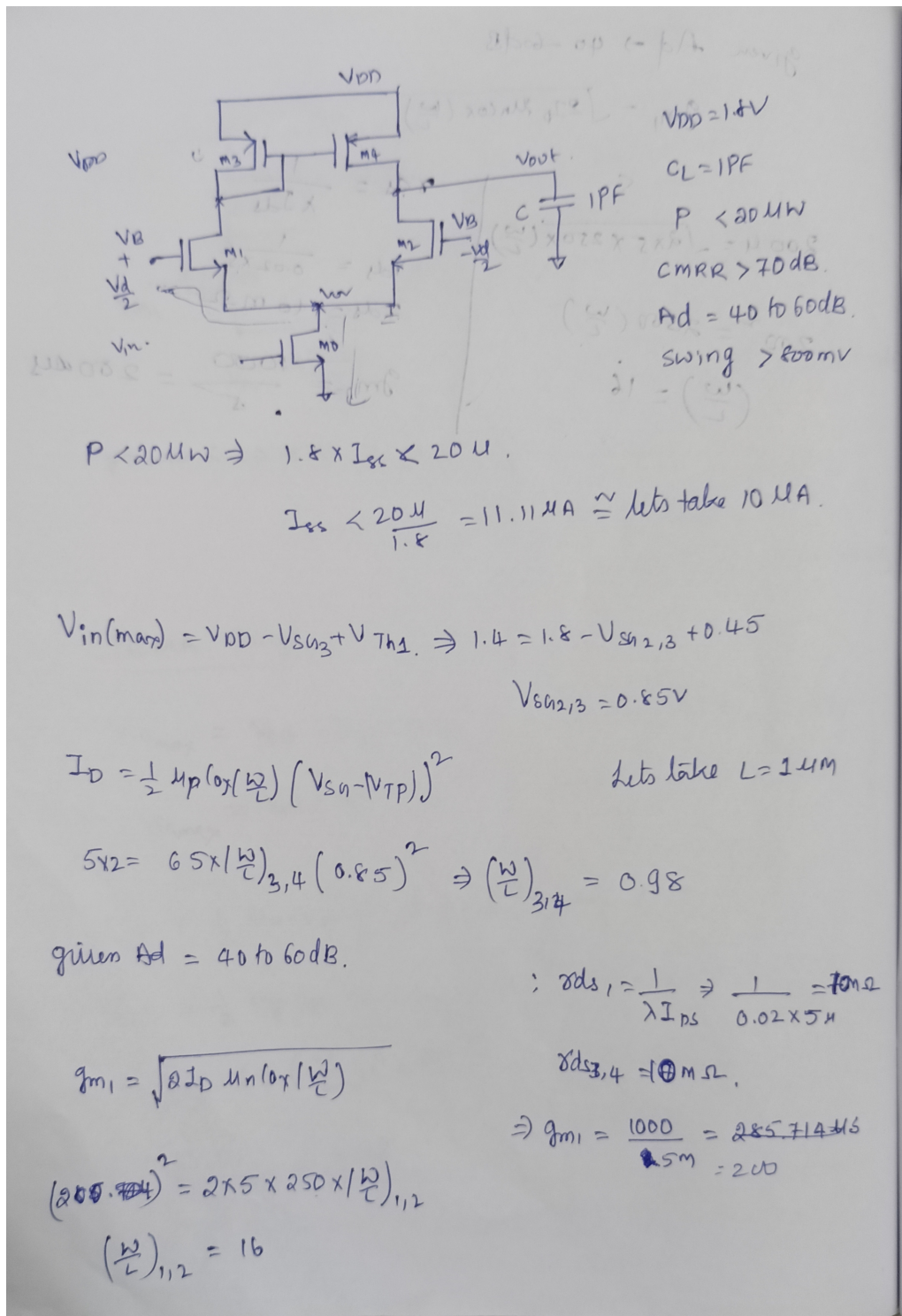


Figure 5: Analysis of the circuit

$$\begin{aligned}
 V_I(\text{min}) &= V_{hS1} + V_{DS0} \\
 &= V_h - V_S + V_D \Rightarrow 0.8 = 0.65 + V_{DS} \\
 V_{DS} &= V_{OV} = 0.15 \text{ V.}
 \end{aligned}$$

$$\Rightarrow \text{not found } \sqrt{\frac{2I_D}{\mu_n C_{ox} \left(\frac{W}{L}\right)}} + V_{TN} = V_{Sg1}$$

$$\Rightarrow \sqrt{\frac{10}{250 \times 16}} + 0.45 = V_{Sg1} \Rightarrow V_{Sg1} = 0.5 \text{ V.}$$

$$\begin{aligned}
 10 &= \frac{1}{2} \times 250 \times \left(\frac{W}{L}\right)_{\text{low}} \times (0.15)^2 \\
 &= \frac{20}{250 \times 0.15^2} = 3.55.
 \end{aligned}$$

$$\left. \begin{aligned}
 r_{ds} &= \frac{1}{\lambda I_{Ds}} = \frac{1}{0.02 \times 54} \\
 r_{ds} &= 7 \text{ m}\Omega \\
 g_{m1} &= \frac{1000}{35} = 28.
 \end{aligned} \right\}$$

Figure 6: Analysis of the circuit